

WEARABLE SAFETY SYSTEM FOR COAL MINES

A PROJECT REPORT

Submitted by

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ABSTRACT

Coal mining is one of the most hazardous industries, where workers are exposed to dangerous environmental conditions such as toxic gases and extreme temperatures. Ensuring the safety and well-being of workers in such environments is a major challenge. This project proposes a Smart Safety System for Coal Mines that integrates Internet of Things (IoT) and wireless communication technologies to provide real-time monitoring and improved safety. The proposed system incorporates multiple sensors, including the MAX30102 sensor for heart rate monitoring, DHT22 sensor for temperature measurement, and gas sensors for detecting harmful gases. These sensors are integrated into a wearable safety system to continuously monitor the health condition of workers and environmental parameters. The collected data is transmitted using LoRa communication from the transmitting node to the receiving node. At the receiver side, an ESP8266 microcontroller acts as a gateway by connecting to a Wi-Fi network and processing the received data. The processed data is then transmitted to a web-based dashboard developed using React JS for real-time visualization. The dashboard displays parameters such as heart rate, temperature, and gas concentration using graphical components, enabling continuous monitoring. In addition to sensor-based monitoring, an attendance monitoring system using computer vision techniques was proposed as part of the system design. However, this module was not fully integrated into the final implementation. Bluetooth Low Energy (BLE) tags are used to verify the presence of safety equipment, ensuring compliance with safety protocols.

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LIST OF ABBREVIATION

IoT	Internet of Things
ESP8266	Espress if Systems Wi-Fi Microcontroller
LoRa	Long Range Communication
MQTT	Message Queuing Telemetry Transport
DHT22	Digital Humidity and Temperature Sensor
MAX30102	Pulse Oximeter and Heart Rate Sensor
CO	Carbon Monoxide
CH ₄	Methane
SPO ₂	Peripheral Capillary Oxygen Saturation
BPM	Beats Per Minute
WSN	Wireless Sensor Network
RF	Radio Frequency
UI	User Interface
IDE	Integrated Development Environment
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
GSM	Global System for Mobile Communication
GPS	Global Positioning System
API	Application Programming Interface
PCB	Printed Circuit Board
LCD	Liquid Crystal Display
LED	Light Emitting Diode
PSU	Power Supply Unit

CHAPTER 1

INTRODUCTION

1.1 Introduction

Coal mining is one of the most essential industries contributing to energy production and industrial growth. However, it is also one of the most hazardous working environments due to the presence of toxic gases, high temperature, humidity, and lack of proper ventilation. These factors pose serious threats to the safety and health of miners working underground.

In recent years, advancements in Internet of Things (IoT) technology have enabled the development of smart monitoring systems that can continuously track environmental and health parameters in real time. This project focuses on designing an IoT-based coal mine safety monitoring system that ensures the safety of miners by detecting hazardous conditions and providing timely alerts.

The system integrates multiple sensors such as gas sensors, temperature sensors, and health monitoring sensors to collect real-time data. This data is transmitted using communication technologies like LoRa and processed through microcontrollers such as ESP8266. The processed data is then sent to a cloud platform for visualization and monitoring.

Additionally, the system incorporates computer vision techniques using OpenCV for attendance and monitoring purposes. The overall system enhances safety, reduces human risk, and improves operational efficiency in coal mines.

1.2 Problem Statement

Coal mining environments are highly dangerous due to the presence of harmful gases such as methane and carbon monoxide, which can lead to

explosions and suffocation. Traditional monitoring systems are often manual and fail to provide real-time data, resulting in delayed responses during emergencies.

Moreover, the lack of continuous health monitoring of miners and inefficient communication systems further increase the risk of accidents. There is a need for an integrated system that can monitor environmental conditions, track miners' health, and provide instant alerts to prevent disasters.

1.3 Proposed Method and Explanation

Wearable Safety System for Coal Mines operates as an integrated ecosystem that bridges the gap between environmental sensing, physiological monitoring, and industrial compliance. At its core, the system utilizes a belt-mounted wearable node powered by an ESP32 microcontroller, which performs real-time edge processing on data gathered from a suite of gas, thermal, and pulse sensors. By relocating the hardware from a traditional helmet to a belt-mounted form factor, the system improves the detection of dense hazardous gases and reduces user fatigue while maintaining PPE compliance through a BLE-based "electronic leash" to the helmet.

To overcome the high attenuation of deep-shaft rock layers, the system employs a 433M Hz LoRa link, ensuring that critical telemetry—such as toxic gas thresholds or abnormal heart rates—is successfully transmitted to a surface-level MQTT dashboard. This multi-layered approach is finalized by a computer vision layer at the mine entrance, which utilizes YOLO-based detection to verify that all personnel are fully equipped with safety vests and helmets before entry, creating a fail-safe environment that combines hardware reliability with intelligent automated supervision.

1.4 Significance of the Study

This project plays a crucial role in improving the safety standards in coal mining industries. By implementing real-time monitoring and alert systems, it reduces the chances of accidents and ensures the well-being of miners.

The integration of IoT and communication technologies provides a reliable and efficient solution for monitoring hazardous environments. It also helps in minimizing human intervention and enables quick decision-making during emergencies.

1.5 Scope of the Study

The scope of this project includes the development of a smart safety system that monitors environmental and health parameters in coal mines. The system focuses on real-time data acquisition, wireless communication, and cloud-based monitoring.

However, the project is limited to prototype-level implementation and does not cover large-scale industrial deployment. Future improvements can include advanced analytics and automation features.

1.6 Overview of the System

The proposed system consists of sensors, microcontrollers, communication modules, and a cloud platform. Sensors collect data related to gas concentration, temperature, humidity, and health parameters.

This data is processed by the ESP8266 microcontroller and transmitted using LoRa communication technology. The data is then sent to a cloud platform

where it is visualized through a dashboard. Alerts are generated when abnormal conditions are detected.



Fig 1.1 System overview

1.7 System Components

The system consists of both hardware and software components working together to ensure safety and monitoring.

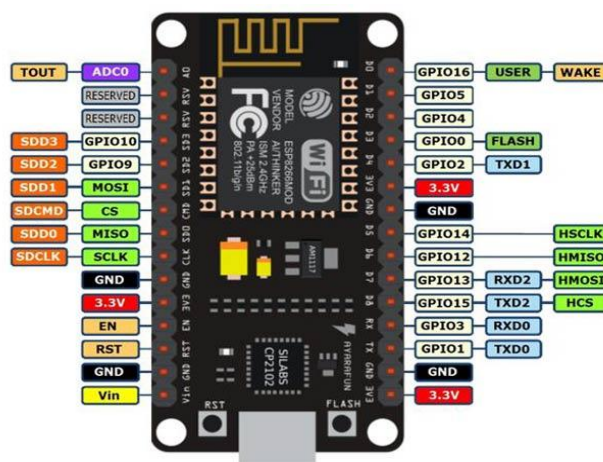


Fig 1.2 ESP8266

1.7.1 Health Monitoring Sensors

Health monitoring sensors such as the MAX30102 are used to measure parameters like heart rate and oxygen levels of miners. These sensors help in tracking the health condition of workers in real time. Monitoring these vitals helps to contribute the whole dedication of the workers towards their work by keeping them healthy and safer from the hazardous environment of the coal mines.

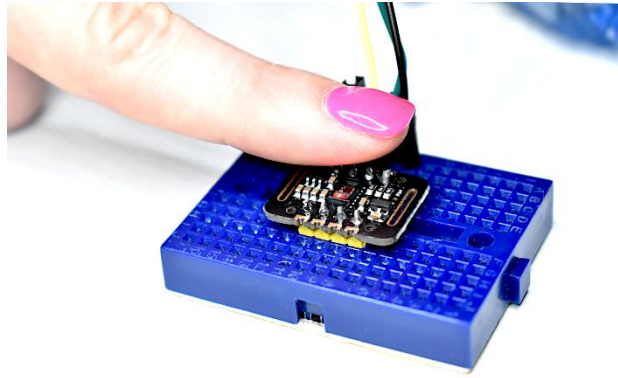


Fig 1.3 MAX30102 Sensor

1.7.2 Environmental Sensors

Environmental sensors include gas sensors and temperature sensors (DHT22). These sensors detect hazardous conditions in the mining environment. The detect in case of emergencies and transmit data's for a better communication within devices.

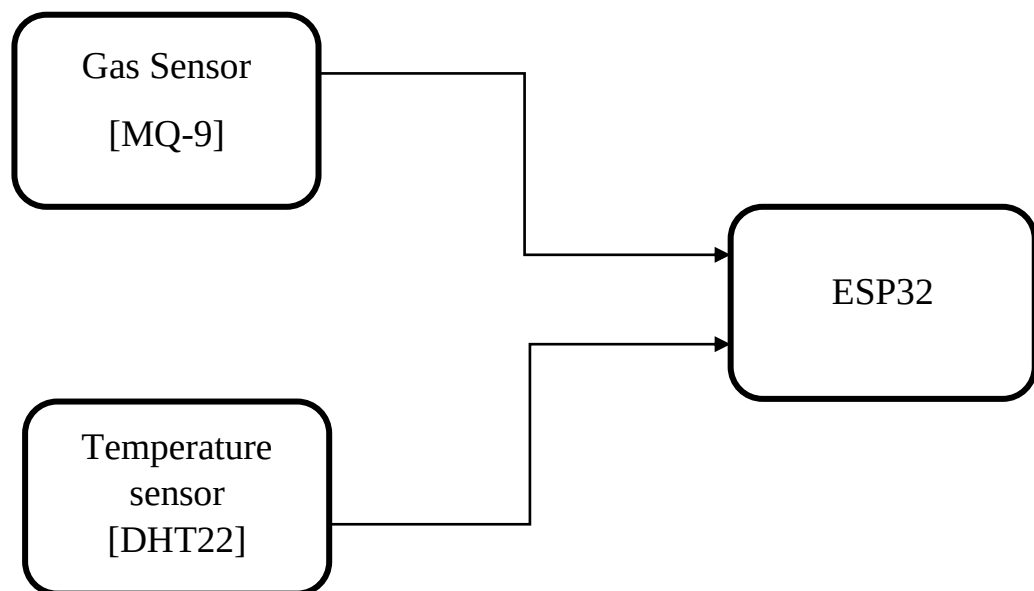


Fig 1.4 Environmental Sensors

1.8 Communication Technologies

The system uses LoRa (Long Range) communication technology for transmitting data over long distances with low power consumption. Additionally, MQTT protocol is used for efficient data communication between devices and the cloud platform. The cloud platform collects the data's and shows them up over a manually created dashboard. The dashboard gives way to access and monitor the workers in a safe and efficient way.

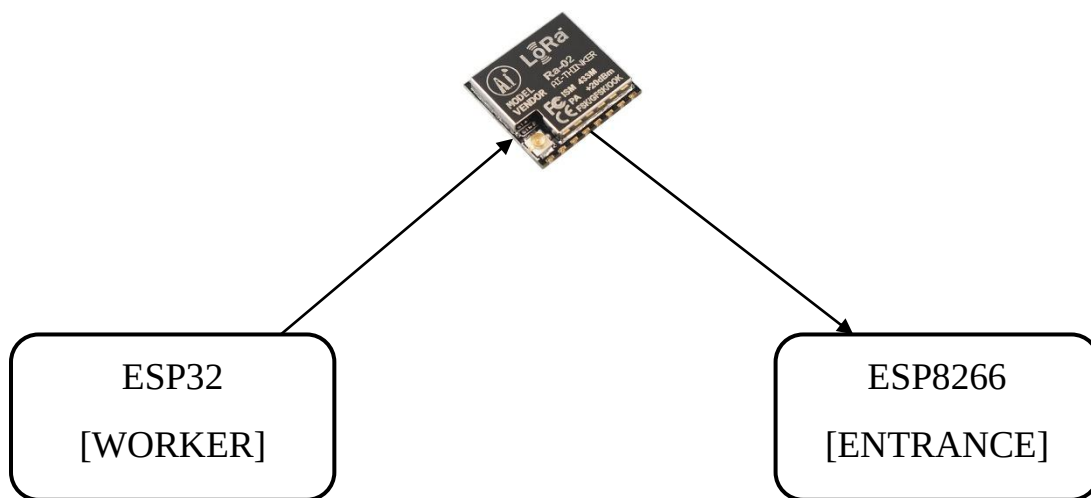


Fig 1.5 Communication Technologies

1.9 Organization of the Report

This report is organized into five chapters. Chapter 1 presents the introduction, problem statement, objectives, and system overview. Chapter 2 discusses the literature survey and existing systems. Chapter 3 describes the proposed system and its advantages. Chapter 4 explains the system design, hardware and software implementation. The Chapter 5 presents the results, discussion, conclusion, and future scope of the project.

SUMMARY

The coal mining industry plays a vital role in supporting energy production and industrial development, but it is also one of the most hazardous working environments. Underground mines are prone to dangerous conditions such as the presence of toxic gases like methane and carbon monoxide, high temperature, humidity, and poor ventilation. These factors pose serious risks to miners, often leading to accidents, health issues, and even loss of life. Ensuring the safety of workers in such environments is therefore a critical concern.

Traditional safety monitoring methods in coal mines mainly rely on manual inspection and basic monitoring systems, which are not capable of providing real-time data. This lack of continuous monitoring leads to delayed detection of hazardous conditions and reduces the effectiveness of emergency response measures. As a result, there is a growing need for an advanced and automated system that can monitor environmental and health parameters continuously and provide timely alerts.

To overcome these challenges, an IoT-based coal mine safety monitoring system is proposed. This system integrates multiple sensors to monitor key environmental parameters such as gas concentration, temperature, and humidity, along with health-related parameters of miners. The collected data is processed using a microcontroller like ESP8266 and transmitted through communication technologies such as LoRa, which enables long-range and low-power data transmission suitable for underground conditions.

The system also includes a cloud-based platform where data is stored, analyzed, and displayed through an interactive dashboard, allowing real-time monitoring and quick decision-making. In addition, advanced features such as

health monitoring and computer vision-based attendance tracking further enhance the overall safety and management of mining operations.

The primary objective of the system is to ensure continuous monitoring, early detection of hazardous conditions, and immediate alert generation to prevent accidents. By integrating modern technologies such as IoT, wireless communication, and data visualization, the system aims to improve safety standards, reduce human risk, and enhance operational efficiency in coal mining environments.

Overall, the proposed approach provides a reliable and efficient solution for addressing the challenges of coal mine safety and highlights the importance of adopting smart technologies in hazardous industrial applications.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Manohara K M, Kiran Kumar, and Ramesh Babu, “IoT Based Coal Mine Safety Monitoring and Alerting System,” IEEE, 2022.

The paper presents a comprehensive approach towards improving safety in underground coal mines using Internet of Things (IoT) technology. Coal mining environments are highly hazardous due to the presence of toxic gases such as methane and carbon monoxide, along with extreme temperature and humidity conditions. Traditional monitoring systems often fail to provide real-time information, which increases the risk of accidents. To overcome these challenges, the authors propose an IoT-based system that enables continuous monitoring of environmental parameters and ensures timely alerts in case of dangerous situations.

The proposed system consists of multiple sensors integrated into a single monitoring unit. Gas sensors are used to detect the presence of harmful gases, while temperature and humidity sensors measure environmental conditions inside the mine. These sensors continuously collect data and send it to a microcontroller, which processes the information and determines whether the conditions are safe or hazardous. The system is designed in such a way that it can detect abnormal variations in environmental parameters at an early stage, thereby preventing major accidents.

One of the key features of the system is its ability to provide real-time data transmission. The processed data is sent to a central monitoring station using wireless communication technologies. This enables supervisors to monitor the conditions inside the mine remotely without physically entering hazardous areas.

In case any parameter exceeds the predefined threshold limit, the system automatically generates alerts in the form of alarms or notifications. This helps in taking immediate action and evacuating workers from dangerous zones.

The authors also highlight the importance of automation in improving mining safety. By reducing the dependency on manual inspection, the system minimizes human errors and ensures continuous monitoring. The use of IoT technology makes the system scalable and flexible, allowing it to be deployed in different mining environments. Additionally, the system is cost-effective compared to traditional wired monitoring systems, making it suitable for small and medium-scale mining operations.

Furthermore, the paper discusses the advantages of integrating multiple sensors into a single platform. This approach provides a more comprehensive understanding of the mining environment and enables better decision-making. The system also supports data logging, which can be used for future analysis and improving safety measures. However, the authors mention that the system has certain limitations, such as dependency on network connectivity and lack of advanced predictive capabilities.

In conclusion, the paper demonstrates that IoT-based monitoring systems can significantly enhance safety in coal mines by providing real-time data, early hazard detection, and efficient communication. The proposed system offers a reliable and practical solution for reducing accidents and improving the overall working conditions in mining environments. However, further improvements can be made by incorporating advanced technologies such as machine learning and health monitoring systems to create a more comprehensive safety solution.

P. P. Kumar, S. R. Patil, “Development of LoRa Communication System for Underground Coal Mines,” IEEE Access, 2023.

This paper focuses on addressing the communication challenges in underground coal mines by developing a LoRa-based communication system. Communication is a critical aspect of mining safety, as it enables data transmission and coordination between workers and control units. Traditional communication systems such as wired networks, Wi-Fi, and ZigBee face limitations in underground environments due to signal attenuation, interference, and limited range.

The authors propose a LoRa-based communication system that provides long-range connectivity with low power consumption. The system is designed to support reliable data transmission from sensor nodes to a central monitoring station. The use of LoRa technology ensures that data can be transmitted over several kilometers, even in challenging conditions.

The paper also discusses the design and implementation of the communication system, including network architecture and performance analysis. The authors demonstrate that LoRa provides better coverage and reliability compared to other wireless technologies. This makes it highly suitable for underground mining applications.

One of the key advantages of the system is its scalability. It can be extended to cover large mining areas by adding more sensor nodes. Additionally, the low power consumption of LoRa devices reduces the need for frequent battery replacement, which is a major advantage in underground environments.

However, the system has certain limitations. The data transmission rate is relatively low, which may not be suitable for applications requiring high-speed

data transfer. Additionally, proper network planning is required to ensure optimal performance.

Overall, the paper highlights the importance of reliable communication in mining safety and demonstrates that LoRa technology provides an effective solution for underground environments.

Harshada R. Rokade, Prof. S. R. Suralkar, “IoT Based Smart Helmet for Coal Mine Safety,” IJRASET, 2021.

This paper presents an innovative approach to coal mine safety by introducing a smart helmet integrated with IoT technology. The primary objective of the proposed system is to enhance the safety of miners by continuously monitoring both environmental conditions and the physical status of workers. Coal mines are highly hazardous environments where workers are exposed to toxic gases, high temperatures, and physical risks such as falls and injuries. Traditional safety systems mainly focus on environmental monitoring and often neglect the health and safety of individual workers. To address this gap, the authors propose a wearable solution that combines multiple sensing technologies into a compact and portable device.

The smart helmet is equipped with various sensors, including gas sensors to detect harmful gases such as methane and carbon monoxide, temperature sensors to monitor heat levels, and humidity sensors to analyze air conditions. In addition to environmental monitoring, the system also incorporates an accelerometer to detect sudden movements, falls, or impacts, which can indicate accidents. This dual monitoring capability makes the system more effective compared to traditional solutions that focus only on environmental factors.

The collected data is processed using a microcontroller embedded within the helmet and transmitted wirelessly to a central monitoring system. This allows

supervisors to track the real-time condition of both the environment and the workers. In case of any abnormal situation, such as high gas concentration or sudden fall detection, the system generates alerts to notify the control unit. This enables quick response and immediate rescue actions, thereby reducing the risk of serious injuries or fatalities.

Another significant advantage of the proposed system is its portability and ease of use. Since the sensors are integrated into a helmet, it does not require additional equipment or complex installation. This makes it highly practical for real-world applications. The authors also emphasize that wearable technology can play a crucial role in improving worker safety, especially in hazardous industries like mining.

However, the system has certain limitations. The performance of the helmet depends on battery life, and continuous operation may require frequent recharging or battery replacement. Additionally, wireless communication in underground environments may face signal interference and connectivity issues. Despite these challenges, the proposed system demonstrates a significant improvement in ensuring miner safety by combining environmental and personal monitoring.

In conclusion, this paper highlights the importance of integrating wearable IoT devices in mining safety systems. The smart helmet provides a comprehensive solution by monitoring both environmental conditions and worker activities, thereby enhancing overall safety and efficiency in coal mining operations.

Wasim Akram, “IoT Based Coal Mine Monitoring System Using LoRa Cloud,” SSRN, 2022.

This paper focuses on the development of a coal mine monitoring system using IoT technology combined with LoRa communication. One of the major challenges in underground mining is establishing reliable communication, as traditional wired systems are difficult to install and maintain. Wireless technologies like Wi-Fi and ZigBee have limited range and are not suitable for deep underground environments. To overcome these limitations, the authors propose the use of LoRa technology, which provides long-range communication with low power consumption.

The proposed system consists of multiple sensors that monitor environmental parameters such as temperature, humidity, gas concentration, and smoke levels. These sensors are connected to a microcontroller, which collects and processes the data. The processed data is then transmitted using LoRa modules to a remote monitoring station. The use of LoRa ensures that data can be transmitted over long distances, even in challenging underground conditions.

One of the key features of the system is its integration with cloud platforms. The collected data is uploaded to a cloud server, where it can be accessed and analyzed in real time. This allows supervisors to monitor the conditions of the mine remotely and take necessary actions in case of emergencies. The cloud-based approach also enables data storage and analysis, which can be used for improving safety measures and predicting potential hazards.

The authors highlight that the use of LoRa technology significantly reduces the complexity and cost of the system. Unlike wired systems, it does not require extensive infrastructure, making it easier to deploy and maintain. Additionally,

the low power consumption of LoRa devices ensures longer battery life, which is crucial for underground applications.

Despite its advantages, the system has certain limitations. The data transmission rate of LoRa is relatively low compared to other communication technologies, which may affect real-time performance in some scenarios. Additionally, the system relies on proper network configuration to ensure reliable communication.

Overall, the paper demonstrates that combining IoT with LoRa communication provides an effective solution for coal mine safety monitoring. The system enhances reliability, reduces cost, and improves safety by enabling real-time monitoring and long-distance communication.

ShakunthalaMasi, R. Ramesh, “IoT Based Coal Mine Safety Monitoring and Controlling,” 2021.

This paper presents an automated safety monitoring and control system for coal mines using IoT technology. The main objective of the system is to reduce accidents by continuously monitoring environmental conditions and providing timely alerts. The authors emphasize that traditional safety systems are not sufficient to handle the dynamic and hazardous nature of mining environments, and there is a need for intelligent systems that can operate autonomously.

The proposed system uses a combination of sensors to detect parameters such as gas concentration, temperature, and humidity. These sensors are connected to a microcontroller, which processes the data and determines whether the conditions are within safe limits. If any parameter exceeds the threshold value, the system triggers an alert to notify workers and supervisors.

In addition to monitoring, the system also includes control mechanisms that can automatically respond to hazardous conditions. For example, ventilation systems can be activated to reduce gas concentration, or alarms can be triggered to warn workers. This automation reduces the need for manual intervention and improves the efficiency of safety management.

The system also supports wireless communication, allowing data to be transmitted to a central monitoring station. This enables remote monitoring and reduces the need for supervisors to enter hazardous areas. The authors highlight that the system is cost-effective and can be easily implemented in small-scale mining operations.

However, the system has some limitations. It does not include advanced features such as predictive analysis or machine learning, which could further enhance safety. Additionally, the system mainly focuses on environmental monitoring and does not address the health conditions of miners.

In conclusion, the paper provides a practical and efficient solution for coal mine safety monitoring. The use of IoT technology enables real-time monitoring and automation, which significantly improves safety and reduces the risk of accidents.

A. H. Ansari, M. Khan, “IoT Based Coal Mine Safety Monitoring and Alerting System,” IEEE, 2021.

This paper presents a comprehensive IoT-based system for monitoring safety conditions in coal mines. The system integrates multiple sensors to detect various hazards such as gas leakage, temperature rise, and water level increase. The authors emphasize that early detection of these hazards is essential for preventing accidents and ensuring the safety of miners.

The proposed system uses a microcontroller to collect data from sensors and transmit it to a monitoring station. One of the unique features of this system is the inclusion of a panic button, which allows miners to send emergency alerts in case of danger. This feature enhances communication and ensures that help can be provided quickly.

The system also uses wireless communication to transmit data, enabling real-time monitoring and reducing the need for manual inspection. The authors highlight that the use of IoT technology improves efficiency and reduces human errors.

However, the system does not include advanced data analysis techniques such as machine learning, which could be used for predicting hazardous conditions. Additionally, the system focuses mainly on environmental monitoring and does not include health monitoring features.

In conclusion, the paper provides a reliable and efficient solution for coal mine safety monitoring. The integration of IoT technology enables real-time monitoring and improves communication, thereby enhancing safety and reducing the risk of accidents.

2.2 LIMITATIONS SUMMARY

From the detailed review of existing coal mine safety monitoring systems, it is observed that although significant advancements have been made using technologies such as Internet of Things (IoT), Wireless Sensor Networks (WSN), LoRa communication, and wearable devices, several limitations still persist.

One of the major limitations is that many systems primarily focus only on environmental monitoring, such as detecting gas concentration, temperature, and humidity, while neglecting the health condition of miners. Although some

systems incorporate wearable devices like smart helmets, they are not widely integrated with the overall monitoring system, resulting in incomplete safety solutions. A comprehensive system should combine both environmental and health monitoring to ensure maximum safety.

Another important limitation is related to communication challenges in underground mining environments. Technologies such as ZigBee and Wi-Fi have limited range and are highly affected by interference and obstacles present in underground tunnels. Although LoRa technology provides better range and reliability, it suffers from low data transmission rates, which may affect real-time performance in certain situations.

Energy consumption is also a critical issue, especially in wireless sensor networks and wearable devices. Continuous monitoring requires sensors and communication modules to operate for long durations, leading to frequent battery depletion. This creates practical challenges in maintaining and replacing batteries in underground conditions.

Cost and complexity are additional concerns. Advanced systems that include robotics, artificial intelligence, or complex communication infrastructures are expensive and difficult to implement, especially in small and medium-scale mining operations. This limits their real-world applicability despite their technological advantages.

Furthermore, most of the existing systems lack predictive capabilities. They are mainly reactive systems that generate alerts only after detecting abnormal conditions. The absence of advanced data analytics and machine learning techniques reduces their ability to predict potential hazards in advance and take preventive measures.

Another limitation is the lack of proper system integration. Many research works focus on individual components such as sensors, communication, or monitoring platforms, but fail to provide a unified system that integrates all these components effectively. This results in inefficiencies and reduced overall performance.

In addition, maintenance and scalability issues are often overlooked. Deploying and maintaining monitoring systems in harsh underground environments is challenging, and many systems are not designed to handle large-scale mining operations efficiently.

Overall, although existing systems have contributed significantly to improving coal mine safety, they still face challenges related to integration, communication, cost, energy efficiency, and predictive capabilities. These limitations highlight the need for a more advanced, reliable, and integrated system that can overcome these challenges and provide a comprehensive solution for coal mine safety monitoring.

CHAPTER 3

PROPOSED SYSTEM

3.1 Introduction

From the analysis of existing coal mine safety monitoring systems, it is evident that although various technologies have been implemented to improve safety, several limitations still exist. Most of the existing systems focus only on environmental monitoring and lack integration with health monitoring, advanced communication, and intelligent data processing. In addition, issues such as limited communication range, high energy consumption, lack of predictive capabilities, and poor system integration reduce the overall efficiency and reliability of these systems.

To overcome these challenges, a more advanced and integrated solution is required that combines environmental monitoring, health monitoring, reliable communication, and real-time data analysis. The proposed system aims to address these limitations by designing a smart coal mine safety monitoring system using Internet of Things (IoT) technology.

The proposed system integrates multiple sensors, including gas sensors, temperature sensors, humidity sensors, and health monitoring sensors, to continuously monitor both environmental and human conditions inside the mine. The system uses a microcontroller such as ESP8266 for processing sensor data and LoRa communication technology for long-range and low-power data transmission. This ensures reliable communication even in underground environments where traditional communication systems fail.

Furthermore, the system incorporates a cloud-based platform for real-time data visualization and monitoring. The collected data is transmitted to the cloud using communication protocols such as MQTT and displayed through an

interactive dashboard. This allows supervisors to monitor the mine conditions remotely and take immediate action in case of emergencies.

In addition to environmental monitoring, the proposed system also includes advanced features such as health monitoring of miners and computer vision-based attendance tracking using OpenCV. These features provide a more comprehensive safety solution compared to existing systems.

Overall, the proposed system is designed to be cost-effective, reliable, and scalable, making it suitable for real-world mining applications. It aims to improve safety standards, reduce human risk, and enhance operational efficiency in coal mining environments.

3.2 System Overview

The proposed coal mine safety monitoring system is a smart and integrated solution that combines sensing, communication, and data processing technologies to ensure the safety of miners. The system continuously monitors environmental parameters such as gas concentration, temperature, and humidity, along with health parameters of workers.

The system consists of sensor nodes placed inside the mining area, which collect real-time data. These sensor nodes are connected to a microcontroller (ESP8266), which processes the data and transmits it using LoRa communication technology. The use of LoRa ensures long-range communication with low power consumption, making it suitable for underground environments.

The transmitted data is received by a gateway and sent to a cloud platform such as ThingsBoard using MQTT protocol. The cloud platform stores, analyzes, and visualizes the data through a user-friendly dashboard. This enables remote monitoring and quick decision-making.

In case of abnormal conditions, such as high gas concentration or temperature rise, the system generates alerts in the form of buzzer notifications and cloud-based alerts. This helps in taking immediate preventive measures and ensuring the safety of miners.

Additionally, the system includes a computer vision module using OpenCV for attendance monitoring and worker tracking. This enhances management and ensures proper supervision of workers.

Thus, the proposed system provides a complete solution by integrating environmental monitoring, health monitoring, communication, and data visualization into a single platform

3.3 Block Diagram Explanation

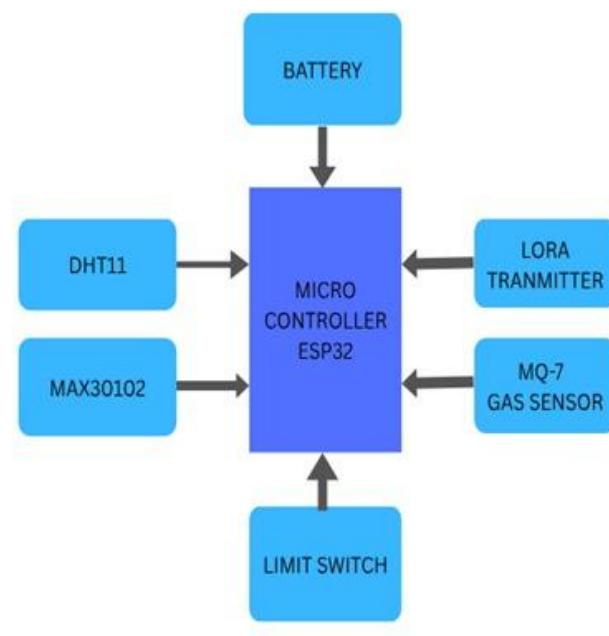


Fig 3.1 Block Diagram of transmitting node

The block diagram of the proposed coal mine safety monitoring system represents the overall architecture and the interaction between different components used in the system. It provides a clear understanding of how data flows from sensors to the cloud and how the system ensures safety through monitoring and alert mechanisms.

The system mainly consists of sensor modules, a processing unit, communication modules, a cloud platform, and output devices. Each block plays a crucial role in the functioning of the system.

At the initial stage, various sensors are deployed inside the coal mine to monitor environmental and health parameters. Gas sensors are used to detect the presence of harmful gases such as methane and carbon monoxide, which are highly dangerous and can lead to explosions. Temperature and humidity sensors, such as DHT22, are used to monitor the environmental conditions inside the mine. In addition, health monitoring sensors like MAX30102 are used to measure vital parameters such as heart rate and oxygen levels of miners. These sensors continuously collect real-time data from the surroundings.

The collected data from all sensors is sent to the microcontroller, which acts as the central processing unit of the system. In this project, ESP8266 (NodeMCU) is used as the microcontroller. It receives input data from sensors, processes it, and checks whether the values are within safe limits. If any parameter exceeds the predefined threshold, the microcontroller identifies it as a hazardous condition.

After processing, the data is transmitted using the LoRa communication module. LoRa is a long-range wireless communication technology that enables data transmission over large distances with low power consumption. This makes it highly suitable for underground coal mines, where traditional communication

systems such as Wi-Fi and ZigBee are not effective. The LoRa module ensures reliable communication between the sensor node and the gateway.

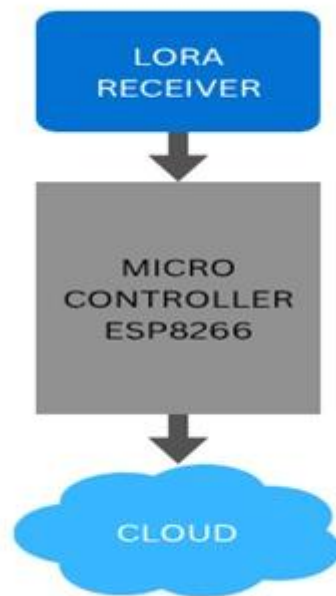


Fig 3.2 Block diagram of receiving node

The transmitted data is then received by a LoRa gateway, which acts as an intermediate device between the sensor nodes and the cloud platform. The gateway forwards the data to the cloud using internet connectivity and communication protocols such as MQTT.

The cloud platform, such as ThingsBoard, is used for data storage, processing, and visualization. It provides a dashboard where all the sensor data is displayed in real time using graphs and indicators. This allows supervisors to monitor the conditions inside the mine remotely. The cloud platform also enables data logging and analysis, which can be useful for future improvements and decision-making.

In addition to monitoring, the system includes an alert mechanism to ensure immediate response during hazardous conditions. When abnormal values are

detected, the system triggers a buzzer or alarm to warn the workers inside the mine. Simultaneously, alerts are sent to the cloud dashboard, enabling authorities to take quick action.

Furthermore, the system integrates a computer vision module using OpenCV for attendance monitoring and worker tracking. A webcam captures images or video, which are processed to identify workers and ensure proper supervision.

Overall, the block diagram illustrates a complete flow of data from sensing to processing, communication, and visualization. Each component is interconnected to ensure continuous monitoring, real-time data transmission, and quick response during emergencies. This integrated approach enhances the safety, reliability, and efficiency of coal mining operations.

3.3.1 Representation

The block diagram of the proposed coal mine safety monitoring system represents the overall structure and interaction between various components involved in the system. It consists of input modules (sensors), a processing unit (ESP8266), communication modules (LoRa), a cloud platform, and output/alert units. The system begins with multiple sensor modules like Gas sensor, Temperature sensor, Health vitals sensors, which are placed in the mining environment to collect real-time data. These sensors are connected to a microcontroller, which processes the data and transmits it through a communication module. The data is then sent to a cloud platform where it is stored, analyzed, and displayed through a dashboard. Alerts are generated whenever abnormal conditions are detected. The sensors mounted on the belt makes this possible.

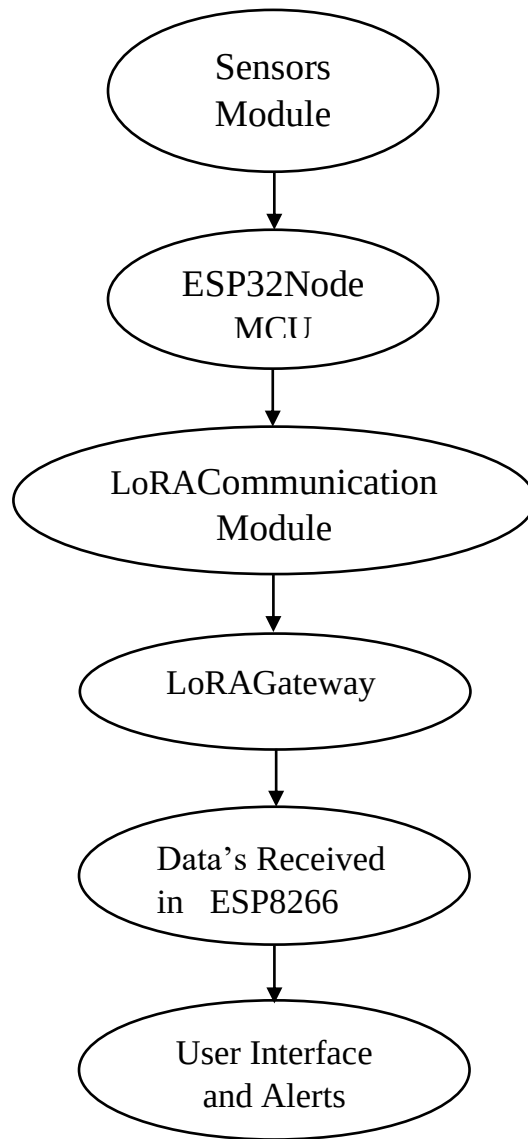


Fig 3.3 Functional flow of proposed system

The block diagram below depicts a clear view about what the project was and what were the main motives of the project to be established. The block diagram gives a clear outline of the project.

3.3.2 Description of Each Block

The first block consists of environmental and health monitoring sensors. Gas sensors are used to detect harmful gases such as methane and carbon monoxide, which are highly dangerous in coal mines. Temperature and humidity sensors (DHT22) monitor environmental conditions, while health sensors

(MAX30102) track vital parameters such as heart rate and oxygen levels of miners.

The second block is the processing unit, which is the ESP8266 (NodeMCU). It acts as the brain of the system by receiving data from sensors, processing it, and determining whether the conditions are safe or hazardous. It also controls the communication and alert mechanisms.

The third block is the communication module, which uses LoRa technology. LoRa enables long-range wireless communication with low power consumption, making it ideal for underground mining environments. It ensures reliable data transmission from sensor nodes to the gateway.

The fourth block is the data collection module, where all the collected data is stored and visualized. The cloud platform provides a user-friendly dashboard for monitoring real-time data and analyzing trends.

The final block includes output and alert systems. These include buzzers and cloud-based notifications that alert workers and supervisors when unsafe conditions are detected. This ensures immediate response and enhances safety.

3.3.3 Data Flow Explanation

The data flow in the system starts from the sensors, which continuously collect environmental and health data. This data is sent to the ESP8266 microcontroller for processing. After processing, the data is transmitted using the LoRa communication module to a gateway.

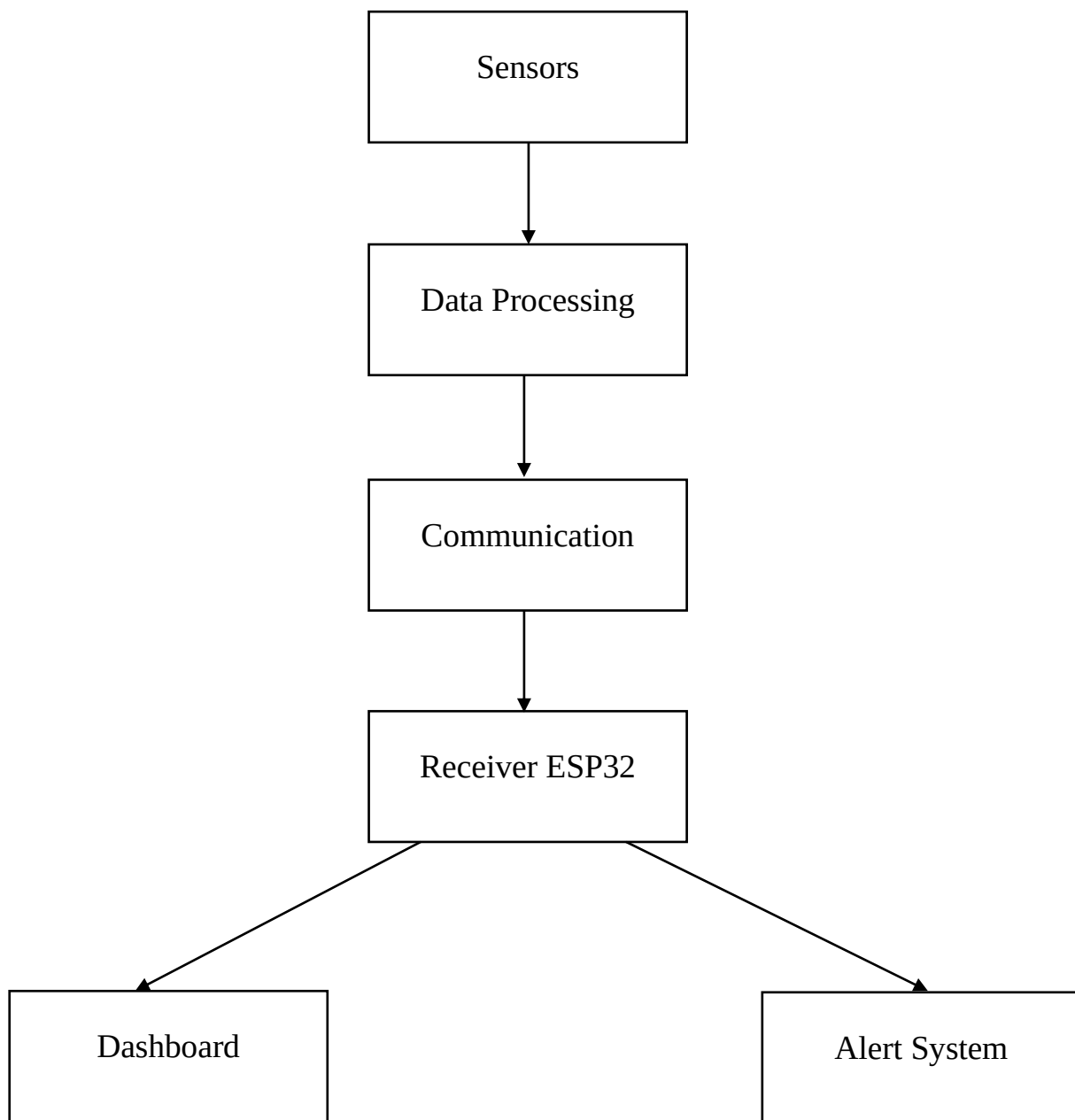


Fig 3.4 Data Flow Diagram

The gateway forwards the data to the cloud platform using internet connectivity and communication protocols such as MQTT. The cloud platform stores and visualizes the data through a dashboard. In case of abnormal conditions, alerts are generated and sent to both local devices (buzzer) and remote users (dashboard notifications).

Thus, the system ensures continuous monitoring, real-time communication, and quick response to hazardous situations.

3.4 Description of Proposed System

The proposed system is an IoT-based coal mine safety monitoring system designed to ensure the safety of miners by continuously monitoring environmental and health parameters in real time. Coal mining environments are highly hazardous due to the presence of toxic gases, extreme temperature conditions, and lack of proper ventilation. To address these challenges, the system integrates multiple technologies such as sensors, wireless communication, cloud computing, and data visualization.

The system primarily consists of sensing units, a processing unit, communication modules, a cloud platform, and an alert mechanism. The sensing unit includes environmental sensors such as gas sensors, temperature sensors (DHT22), and humidity sensors, which are used to detect hazardous conditions inside the mine. Gas sensors play a crucial role in detecting harmful gases such as methane and carbon monoxide, which can cause explosions or suffocation. Temperature and humidity sensors help in monitoring environmental conditions and identifying abnormal changes that may indicate potential risks.

In addition to environmental monitoring, the system also incorporates a health monitoring module using sensors such as MAX30102. This module tracks vital parameters of miners, including heart rate and oxygen levels, ensuring their physical well-being. By integrating both environmental and health monitoring, the system provides a comprehensive safety solution.

The collected data from all sensors is processed using the ESP8266 (NodeMCU) microcontroller, which acts as the central control unit. It continuously analyzes the sensor data and compares it with predefined threshold values to determine whether the conditions are safe or hazardous. The microcontroller also manages communication and alert operations.

For communication, the system uses LoRa technology, which enables long-range wireless data transmission with low power consumption. This makes it highly suitable for underground mining environments where traditional communication systems such as Wi-Fi and ZigBee may fail due to limited range and interference. The LoRa module transmits the processed data to a gateway, which forwards it to a cloud platform using internet connectivity.

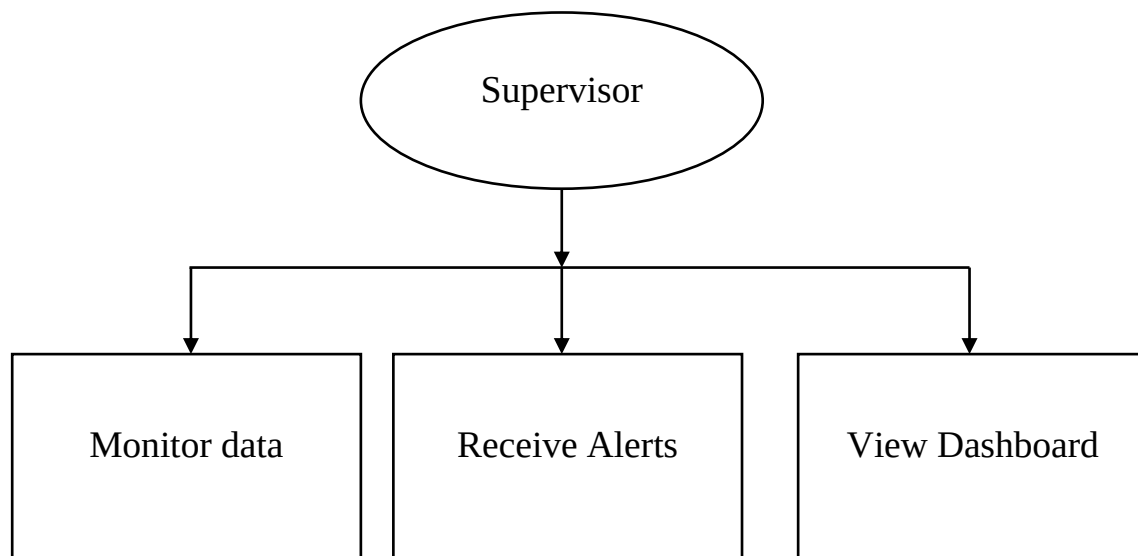


Fig 3.5 Supervising End

The cloud platform, such as Things Board, is used for storing, analyzing, and visualizing the data. It provides a user-friendly dashboard where real-time data is displayed in the form of graphs and indicators. This allows supervisors to monitor mining conditions remotely and make informed decisions.

An alert mechanism is also integrated into the system to ensure immediate response during hazardous situations. When any parameter exceeds the predefined safe limit, the system generates alerts through buzzers and cloud notifications. This enables workers and authorities to take quick action and prevent accidents.

Additionally, the system includes a computer vision module using OpenCV for attendance monitoring and worker tracking. A webcam captures images or video, which are processed to identify workers and ensure proper supervision. This adds an extra layer of safety and management to the system.

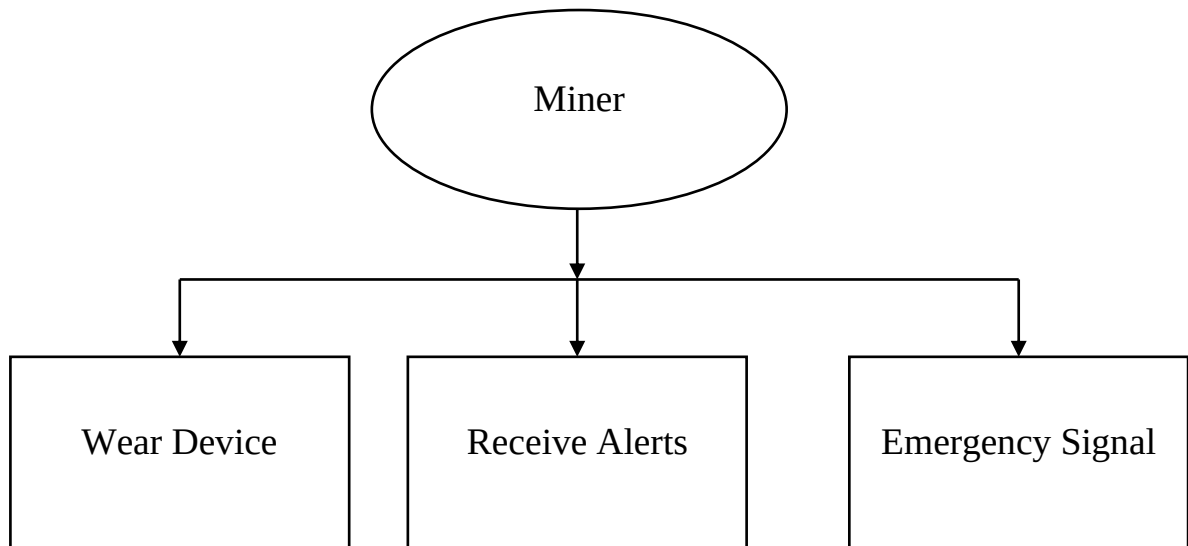


Fig 3.6 Data Transmitter's End

Overall, the proposed system provides a reliable, efficient, and cost-effective solution for coal mine safety by integrating multiple technologies into a single platform. It ensures continuous monitoring, real-time communication, and quick response to hazardous conditions.

3.4.1 Environmental Monitoring Module

The environmental monitoring module is responsible for detecting hazardous conditions inside the coal mine. It uses sensors such as gas sensors to detect toxic gases, temperature sensors to monitor heat levels, and humidity sensors to analyze air conditions. These sensors continuously collect data and help in identifying unsafe conditions at an early stage.

The environmental monitoring module is one of the most critical components of the proposed coal mine safety monitoring system, as it is responsible for continuously observing the surrounding conditions within the mining environment. Coal mines are highly hazardous due to the presence of toxic and combustible gases, extreme temperature variations, and poor ventilation. Therefore, real-time monitoring of these environmental parameters is essential to prevent accidents and ensure the safety of workers.

This module consists of multiple sensors such as gas sensors, temperature sensors, and humidity sensors. The gas sensor plays a vital role in detecting harmful gases such as methane (CH_4), carbon monoxide (CO), and other toxic gases that may accumulate in underground mines. Methane is highly flammable and can lead to explosions if its concentration exceeds safe limits, while carbon monoxide is poisonous and can cause serious health issues or even death. The gas sensor continuously measures the concentration of these gases and sends the data to the microcontroller for processing.

The temperature sensor (DHT22) is used to monitor the temperature levels inside the mine. Sudden increases in temperature may indicate potential fire hazards or equipment malfunction. Similarly, the humidity sensor measures the moisture content in the air, which is important for maintaining a safe and comfortable working environment. High humidity levels can lead to equipment corrosion and affect the performance of electronic devices.

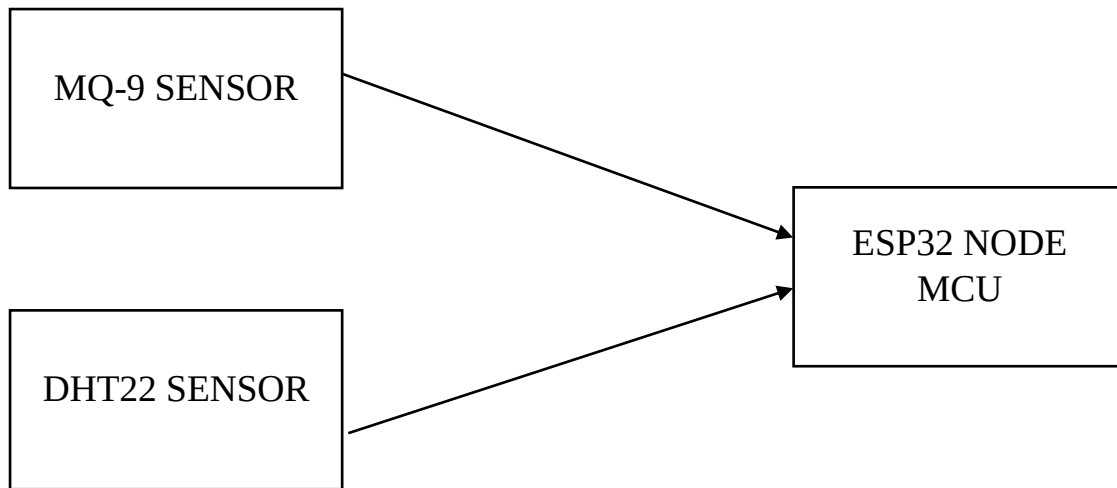


Fig 3.7 Environmental Sensors Setup

All these sensors work together to provide a comprehensive view of the environmental conditions. The data collected is continuously transmitted to the processing unit, where it is analyzed in real time. If any parameter exceeds the predefined threshold values, the system immediately identifies it as a hazardous condition.

The environmental monitoring module ensures early detection of dangerous situations, allowing preventive measures to be taken before accidents occur. This significantly reduces the risk of explosions, suffocation, and other hazards in coal mines, thereby enhancing the overall safety of the mining environment.

3.4.2 Health Monitoring Module

The health monitoring module tracks the physical condition of miners using sensors like MAX30102. It measures parameters such as heart rate and oxygen levels, ensuring that workers are safe and healthy. This module helps in

detecting health-related issues and provides alerts when abnormal conditions are observed.

The health monitoring module is designed to ensure the well-being of miners by continuously tracking their physiological parameters. In coal mining environments, workers are often exposed to extreme conditions such as high temperatures, toxic gases, and physical exertion, which can adversely affect their health. Therefore, monitoring the health status of workers is equally important as monitoring environmental conditions.

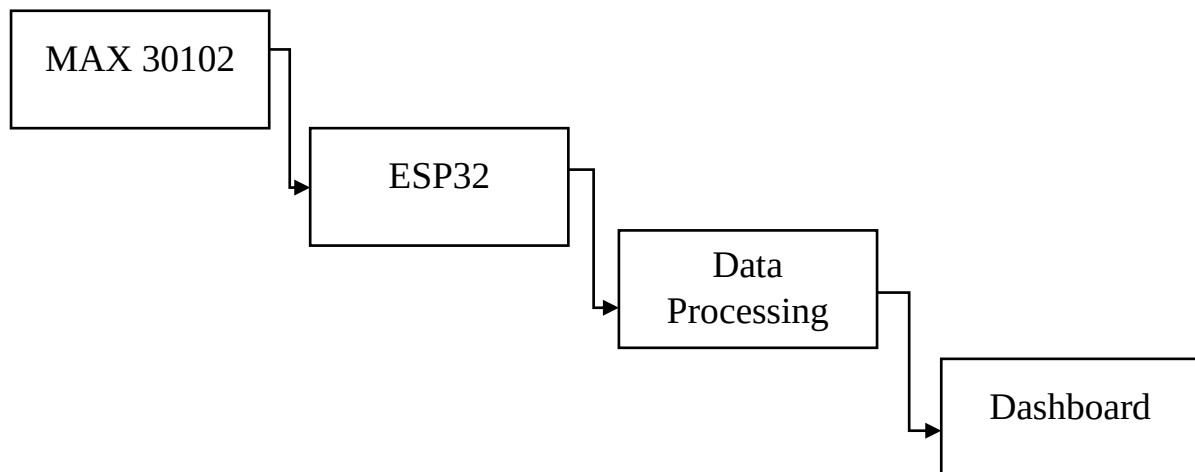


Fig 3.8 Health Data Flow Diagram

This module utilizes sensors such as the MAX30102, which is capable of measuring heart rate and blood oxygen saturation (SpO2) levels. The heart rate sensor helps in detecting abnormal heart conditions such as tachycardia or bradycardia, while the oxygen level sensor ensures that the worker is receiving sufficient oxygen. A sudden drop in oxygen levels may indicate exposure to harmful gases or poor ventilation.

The collected health data is continuously transmitted to the microcontroller, where it is processed and analyzed. The system compares the measured values with predefined normal ranges to determine whether the worker is in a safe condition. If any abnormality is detected, the system generates an alert to notify both the worker and the supervisor.

This module plays a crucial role in preventing health-related emergencies. For example, if a worker experiences fatigue, dizziness, or breathing difficulties, the system can detect these changes early and provide timely alerts. This enables quick medical intervention and reduces the risk of serious health issues.

Additionally, the integration of health monitoring with environmental monitoring provides a holistic approach to safety. By analyzing both environmental and physiological data, the system can make more accurate assessments of potential risks. This ensures that workers are not only protected from external hazards but also from internal health issues.

3.4.3 Communication Module

The communication module is responsible for transmitting data from the sensor nodes to the cloud platform. It uses LoRa technology, which provides long-range communication with low power consumption. This ensures reliable data transmission even in underground environments where other communication technologies fail. In underground mining environments, communication is a major challenge due to the presence of obstacles, interference, and limited connectivity. Therefore, selecting an appropriate communication technology is essential for the effective functioning of the System.

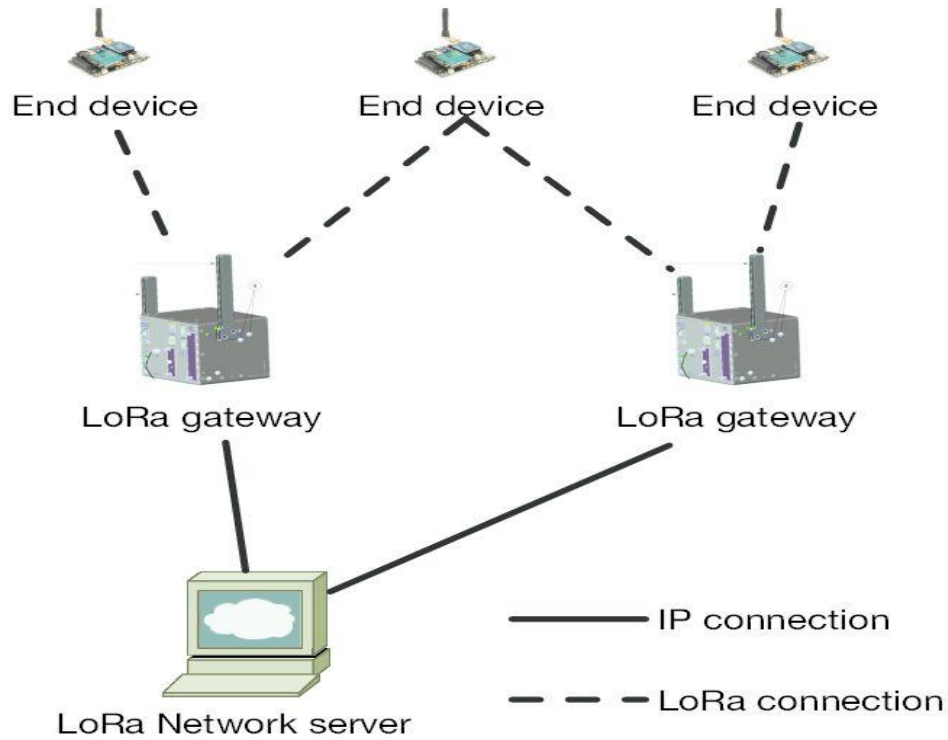


Fig 3.9 LoRA Communication Architecture Diagram

In this proposed system, LoRa (Long Range) communication technology is used due to its ability to provide long-range wireless communication with low power consumption. Unlike traditional communication technologies such as Wi-Fi and ZigBee, which have limited range and are affected by obstacles, LoRa can transmit data over several kilometres, making it suitable for underground environments.

The LoRa module is connected to the ESP8266 microcontroller and is responsible for sending the processed sensor data to a gateway. The gateway acts as an intermediary between the sensor nodes and the cloud platform. It receives data from multiple nodes and forwards it to the cloud using internet connectivity.

One of the key advantages of LoRa communication is its low power consumption, which allows the system to operate for extended periods without

frequent battery replacement. This is particularly important in mining environments where access to power sources may be limited.

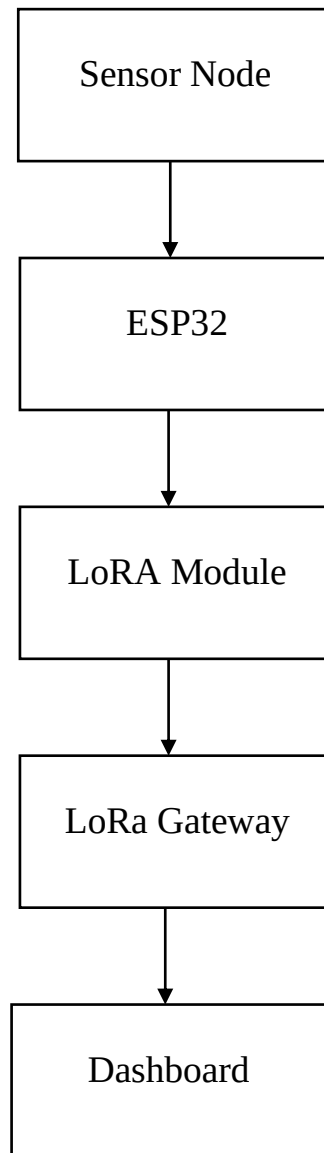


Fig 3.10 LoRa Communication Architecture

The communication module ensures reliable and continuous data transmission, enabling real-time monitoring of the system. It also supports scalability, allowing additional sensor nodes to be added without affecting the overall performance.

Overall, the communication module plays a vital role in connecting different components of the system and ensuring seamless data flow from sensors to the cloud.

3.4.4 Real-Time Data Transmission and Monitoring using MQTT

The cloud integration module stores and processes the collected data. Platforms like ThingsBoard are used to visualize the data through dashboards. This enables remote monitoring and analysis of mining conditions, allowing supervisors to make informed decisions.

The cloud integration module plays a crucial role in the proposed system by enabling real-time data storage, processing, and visualization. It acts as a centralized platform where all the data collected from sensors is transmitted, analyzed, and displayed in an understandable format. In this system, cloud computing is integrated with the Internet of Things (IoT) to provide efficient monitoring and remote accessibility.

The communication between the hardware components and the cloud platform is achieved using the MQTT (Message Queuing Telemetry Transport) protocol. MQTT is a lightweight messaging protocol specifically designed for IoT applications, where bandwidth and power consumption are limited. It follows a publish-subscribe architecture, which makes it highly efficient for real-time communication.

In this architecture, the ESP8266 microcontroller acts as a publisher, which collects data from various sensors and publishes it to specific topics. These topics are managed by an MQTT broker, which acts as an intermediary between the publisher and subscribers. The cloud platform subscribes to these topics and receives the data in real time.

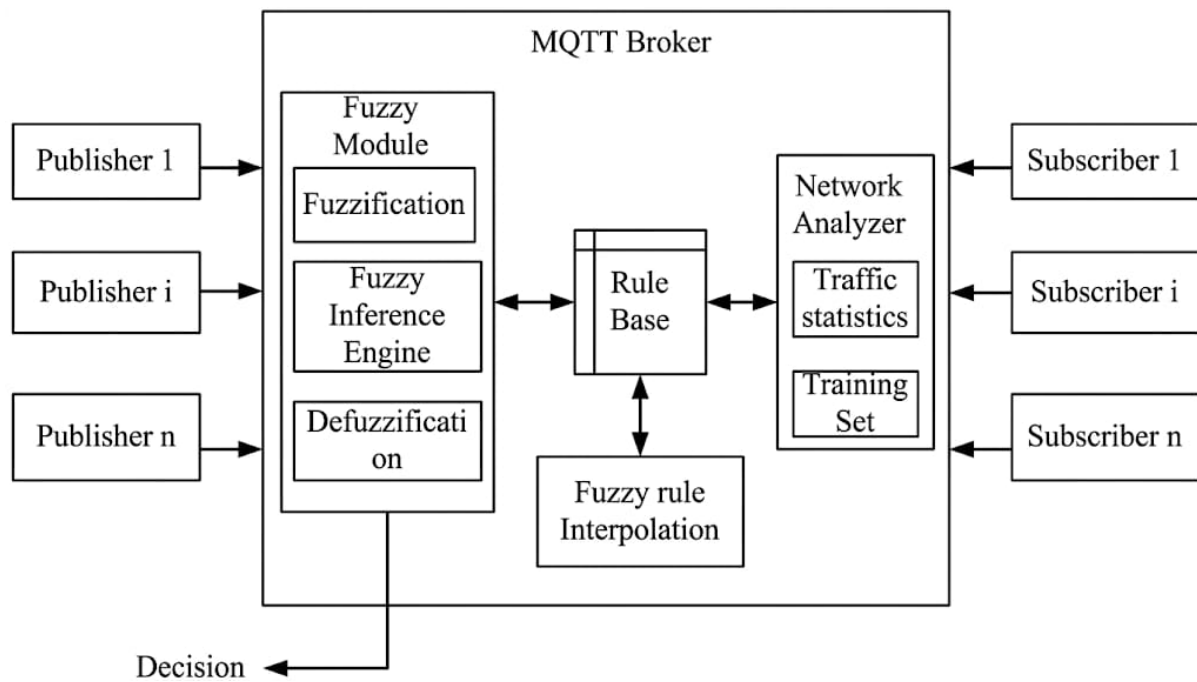


Fig 3.11 MQTT Architecture

The MQTT protocol ensures reliable and efficient data transmission even in low-bandwidth conditions, which is particularly useful in mining environments. It reduces network load and allows multiple devices to communicate simultaneously without direct connections. This improves the scalability and flexibility of the system.

Once the data reaches the cloud platform, it is stored and processed for visualization. The dashboard provides a graphical representation of the data, including sensor readings, worker status, and alert notifications. The use of charts, graphs, and indicators makes it easier for users to understand the system status and take appropriate actions.

The cloud platform also enables remote monitoring, allowing supervisors to access the system from any location using internet-enabled devices. This enhances decision-making and ensures quick response to hazardous situations.

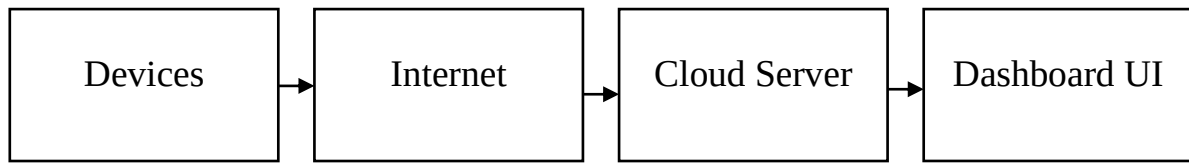


Fig 3.12 Real-Time Data Flow Architecture

In addition to real-time monitoring, the cloud integration module supports data logging and historical analysis. The stored data can be used to identify patterns, predict potential risks, and improve safety measures in the mining environment.

Overall, the integration of MQTT with cloud computing enhances the efficiency, reliability, and scalability of the proposed system. It ensures seamless communication between hardware and software components, making the system more robust and intelligent.

3.4.5 Alert and Safety Mechanism

The alert system plays a crucial role in ensuring safety. When any parameter exceeds the predefined threshold, the system generates alerts in the form of buzzer sounds and cloud notifications. This helps workers and authorities to take immediate action and prevent accidents.

The alert and safety mechanism is a crucial component of the proposed system, as it ensures immediate response during hazardous conditions. The primary objective of this module is to notify workers and supervisors about potential dangers so that timely action can be taken to prevent accidents.

The system continuously monitors sensor data and compares it with predefined threshold values. When any parameter exceeds the safe limit, the

system identifies it as a hazardous condition and triggers an alert. The alert is generated in multiple forms to ensure maximum effectiveness.

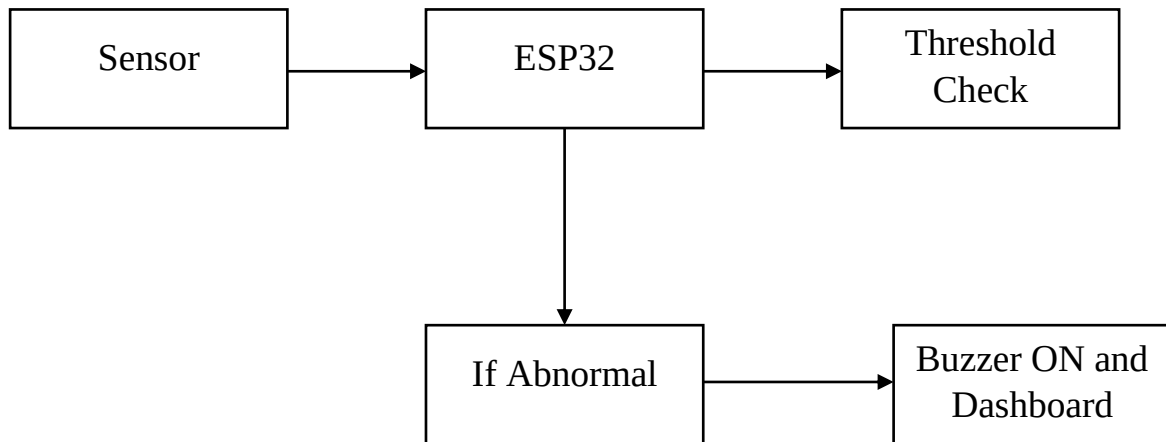


Fig 3.13 Alert Flow Mechanism

Firstly, a buzzer is activated to provide an audible warning to the worker. This immediate alert helps the worker to take quick action, such as evacuating the area or seeking assistance. Secondly, notifications are sent to the cloud dashboard, where supervisors can view the alert in real time. This enables remote monitoring and quick decision-making.

In addition to these alerts, the system can be extended to include SMS or mobile notifications for enhanced communication. This ensures that alerts are received even if the supervisor is not actively monitoring the dashboard.

The alert system is designed to be reliable and responsive, ensuring that no critical situation goes unnoticed. It plays a vital role in preventing accidents and minimizing risks in coal mining environments.

Overall, the alert and safety mechanism enhances the effectiveness of the system by providing timely warnings and enabling quick response to hazardous situations.

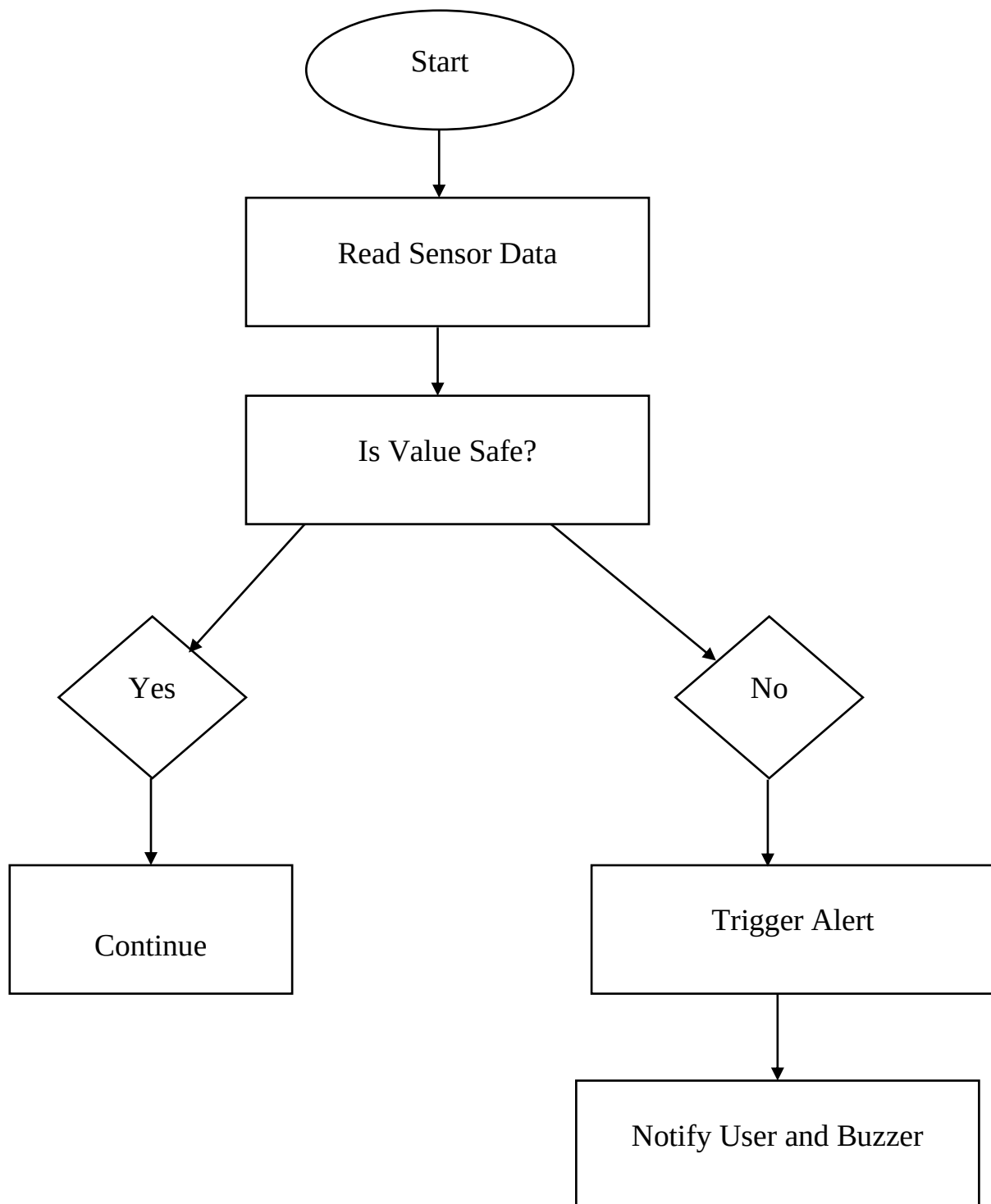


Fig 3.14 Decision Flow Chart

3.5 Advantages of Proposed System

There are various advantages of the system that has been proposed to resolve the problems of the coal mine workers to work peacefully without the threat to their lives. Advantages classify a major part in creating and improvising the system that has been developed.

3.5.1 Technical Advantages

The proposed system offers several technical advantages that make it more effective compared to traditional safety monitoring systems. One of the key advantages is real-time monitoring, which enables continuous tracking of environmental and health parameters. This ensures early detection of hazardous conditions and reduces the risk of accidents.

The use of LoRa communication technology provides long-range data transmission with low power consumption. This makes the system suitable for underground mining environments where other communication technologies are not reliable. The system also integrates multiple sensors into a single platform, allowing comprehensive monitoring of both environmental and human factors.

Another important advantage is cloud-based data storage and visualization. The use of a cloud platform allows remote access to data, enabling supervisors to monitor conditions from anywhere. It also supports data analysis and logging, which can be used for improving safety measures.

The system is scalable and flexible, meaning it can be easily expanded by adding more sensors or modules. This makes it suitable for both small-scale and large-scale mining operations. Additionally, the integration of computer vision enhances monitoring and management capabilities.

3.5.2 Practical Advantages

From a practical perspective, the proposed system significantly improves the safety of miners by providing timely alerts and reducing the chances of accidents. It minimizes the need for manual inspection, thereby reducing human errors and exposure to hazardous conditions.

The system is cost-effective compared to traditional wired monitoring systems, as it reduces installation and maintenance costs. The use of wireless communication eliminates the need for complex infrastructure, making it easier to deploy in underground environments.

Another advantage is ease of use and maintenance. The system is designed to be user-friendly, allowing operators to monitor data through simple dashboards. It also requires minimal maintenance, making it practical for real-world applications.

Furthermore, the system improves operational efficiency by enabling quick decision-making and effective management of mining activities. It ensures better supervision of workers through health monitoring and attendance tracking features.

Overall, the proposed system provides a comprehensive and efficient solution for coal mine safety, combining advanced technology with practical usability to enhance safety and productivity.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents a detailed analysis of the results obtained from the implementation of the proposed coal mine safety monitoring system. The system was designed to ensure the safety of miners by continuously monitoring environmental conditions and health parameters using IoT-based technologies. The results include both hardware and software outputs, demonstrating the effectiveness of the system in real-time monitoring and alert generation.

The primary objective of this chapter is to evaluate the performance of the developed system under different conditions and to verify whether the system meets the intended goals. The discussion also highlights how the integration of sensors, communication modules, and cloud-based platforms contributes to enhancing safety in mining environments.

The proposed system was tested using a prototype model, which simulates real-world mining conditions. The system includes multiple components such as gas sensors, temperature and humidity sensors, health monitoring sensors, a microcontroller (ESP8266), LoRa communication modules, and a cloud-based dashboard. Additionally, computer vision techniques were implemented for worker identification and attendance monitoring.

This chapter provides a comprehensive understanding of how the system operates in real-time, how data is processed and transmitted, and how the results are visualized and analysed. It also discusses the performance of the system in terms of accuracy, reliability, response time, and efficiency.

4.2 Experimental Setup

The experimental setup of the proposed system involves the integration of hardware components into a compact and portable prototype model. The system is designed to resemble a wearable safety device that can be used by miners in real-time conditions. The prototype is enclosed within a box structure and attached to a belt, making it easy to carry and operate.

The internal structure of the device consists of a breadboard setup where all the electronic components are connected. The ESP8266 (NodeMCU) microcontroller serves as the central processing unit, which is responsible for collecting and processing data from various sensors. The sensors used in the system include a gas sensor for detecting harmful gases, a DHT22 sensor for measuring temperature and humidity, and a MAX30102 sensor for monitoring health parameters such as heart rate and oxygen levels.

The system is powered by a rechargeable battery, which ensures continuous operation without interruption. Proper wiring and connections are established between the sensors, microcontroller, and communication modules to ensure smooth data flow.

The LoRa communication module is integrated into the system to enable long-range wireless communication. This module transmits the collected data to a gateway, which further sends the data to a cloud platform using internet connectivity.

The prototype model clearly demonstrates how the system can be implemented in a real mining environment. The compact design ensures portability, while the integration of multiple components provides a comprehensive monitoring solution.

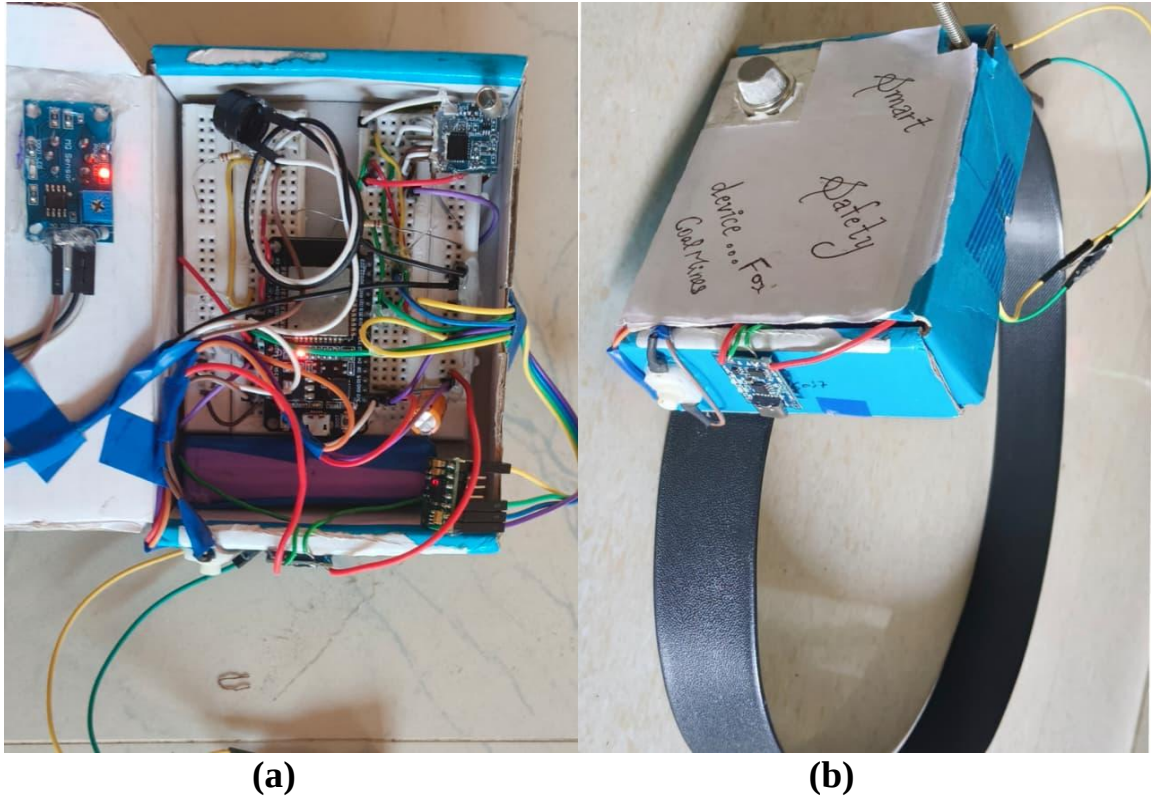


Fig 4.1 Experimental Setup Belt Image

4.3 Output Results

The output results of the system demonstrate its ability to collect, process, and transmit real-time data effectively. The sensors continuously monitor environmental conditions such as gas concentration, temperature, and humidity, as well as health parameters of workers.

During testing, the system successfully detected variations in environmental conditions. For example, when the gas sensor was exposed to gases, the system immediately detected the increase in gas concentration and triggered alerts. Similarly, changes in temperature and humidity were accurately measured and displayed.

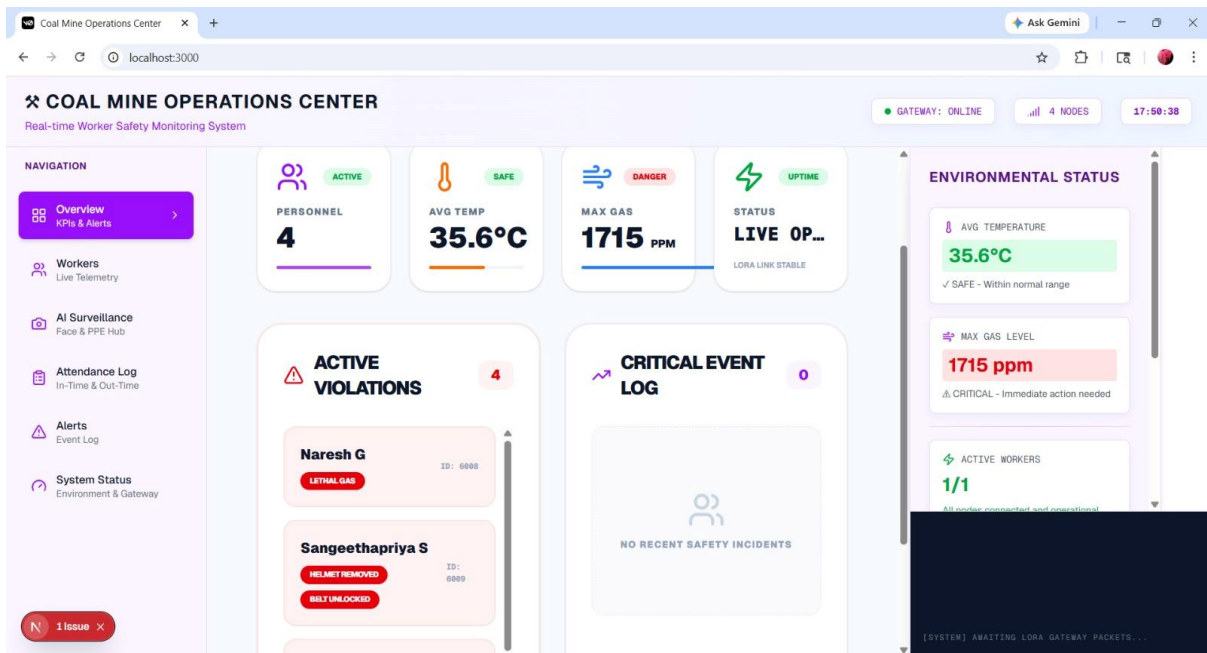


Fig 4.2 Sensor's Outputs in Dashboard

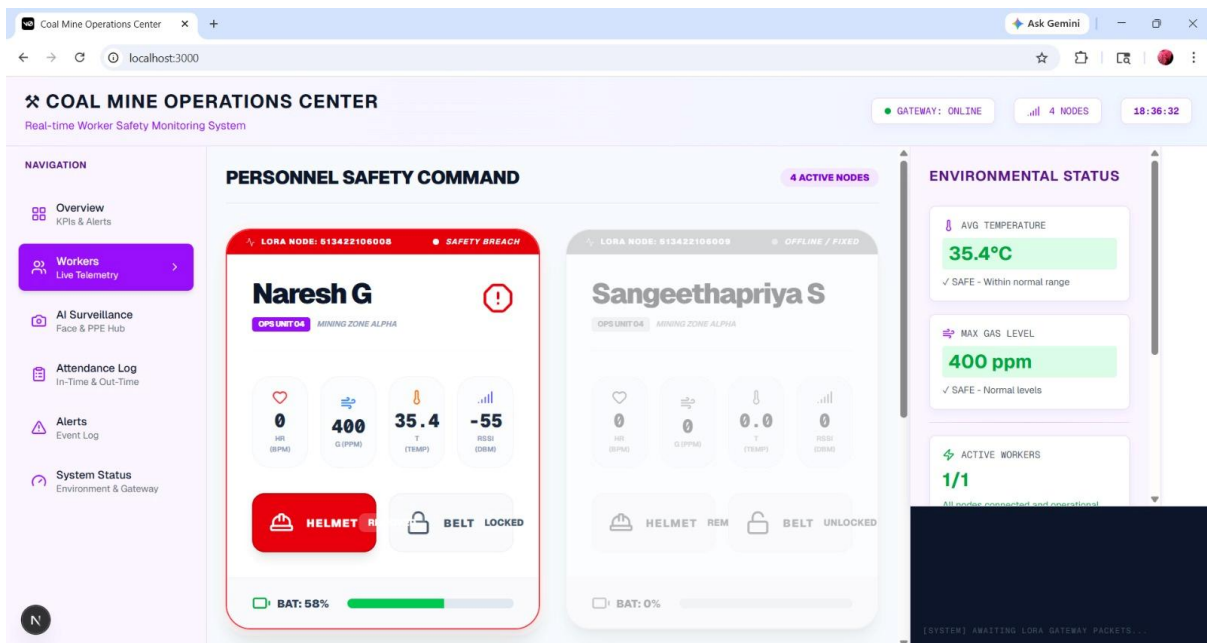


Fig 4.3 Specified worker's Sensor output status

The health monitoring sensor provided real-time readings of heart rate and oxygen levels. These readings were processed by the microcontroller and transmitted to the cloud platform for visualization.

The system also includes an alert mechanism that activates when any parameter exceeds the predefined threshold values. The buzzer produces an audible alert to warn the worker, while notifications are sent to the dashboard for remote monitoring. The results confirm that the system is capable of real-time monitoring and quick response to hazardous conditions, making it suitable for safety-critical applications such as coal mining.

4.4 Dashboard Results

The dashboard plays a crucial role in the system by providing a real-time interface for monitoring and analysis. The dashboard is designed using a cloud-based platform and displays various parameters such as sensor readings, worker status, alerts, and system performance.

One of the key features of the dashboard is the live camera streaming functionality. The system uses a webcam to capture real-time video, which is processed using computer vision techniques. The face recognition module identifies workers and displays their names on the screen. This ensures proper identification and monitoring of workers.

The AI surveillance module enhances safety by detecting the presence of workers and verifying their identity. This helps in preventing unauthorized access and ensures that all workers are accounted for.

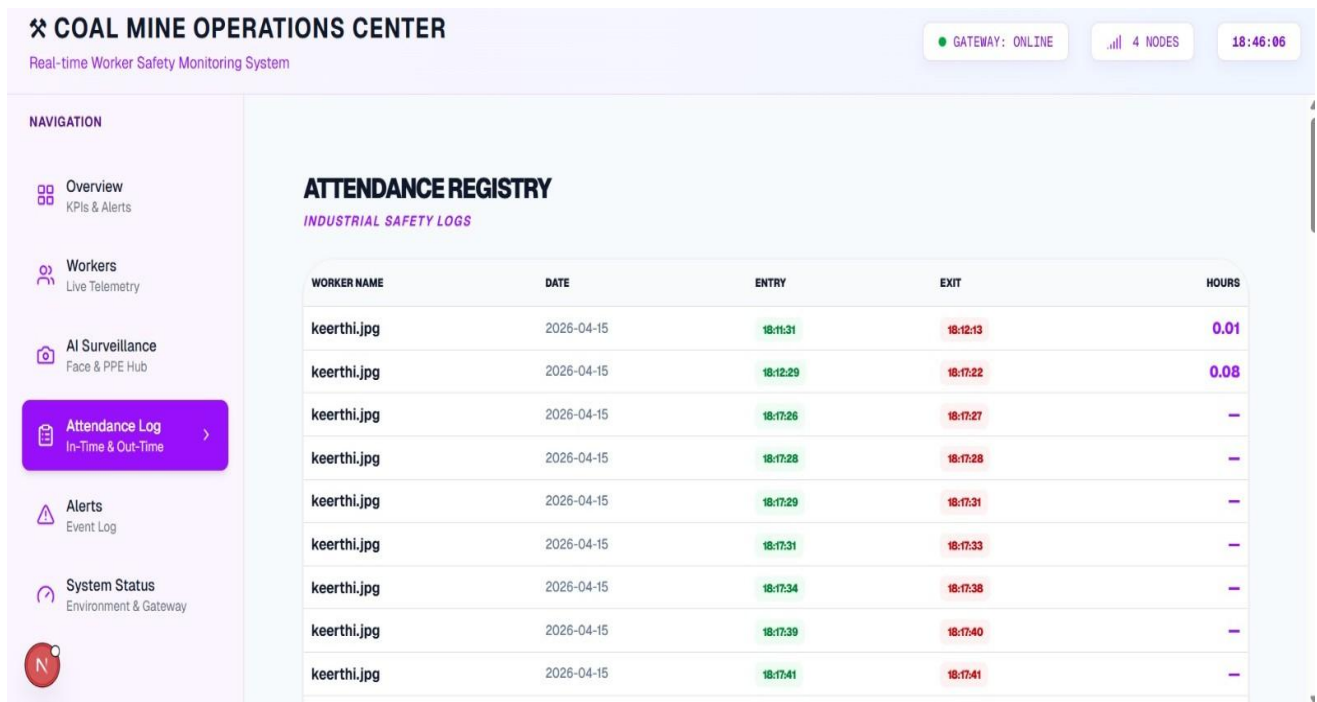


Fig 4.4 Outputs of the attendance of the workers

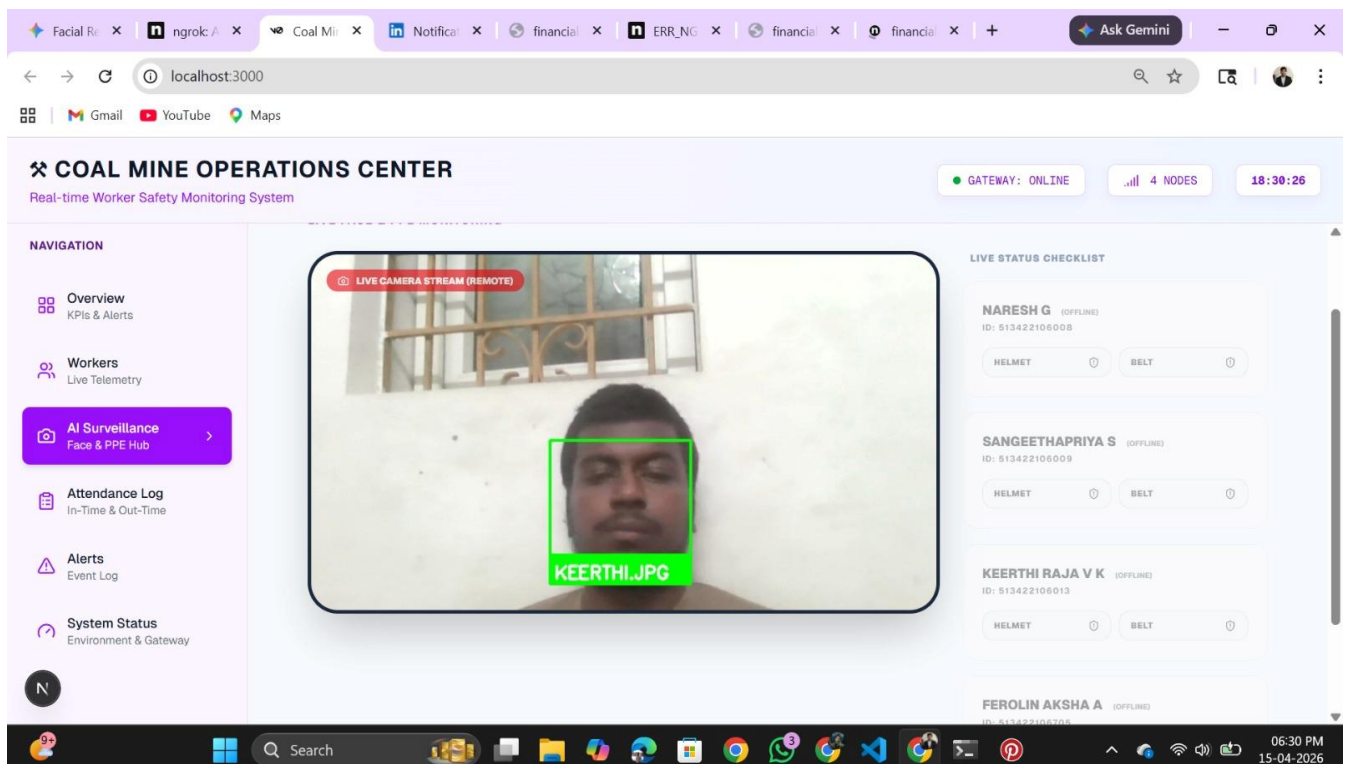


Fig 4.5 Output of Computer Vision attendance and Safety PPE's checking system

Another important feature is the attendance registry, which records the entry and exit times of workers. The system automatically logs the time when a worker is detected by the camera, and updates the attendance record accordingly. This eliminates the need for manual attendance tracking and improves safety.

The dashboard also displays system status indicators such as gateway connectivity, number of active nodes, and current time. These features provide a comprehensive overview of the system and its performance.

The graphical representation of data through charts and indicators makes it easier to analyze trends and identify potential issues. Overall, the dashboard provides a user-friendly and efficient interface for monitoring and control.

4.5 Performance Analysis

The performance of the proposed system was evaluated based on several key parameters, including accuracy, response time, communication efficiency, reliability, and scalability.

The system demonstrated high accuracy in detecting environmental and health parameters. The sensors provided reliable readings, which were processed and transmitted without significant errors. The response time of the system was minimal, allowing quick detection and alert generation.

The use of LoRa communication technology ensured efficient data transmission over long distances. The system was able to maintain stable communication even in conditions where traditional communication methods may fail. This makes the system highly suitable for underground mining environments.

The reliability of the system was observed to be high, as it was able to operate continuously without major interruptions. The integration of hardware and software components was smooth, resulting in consistent performance.

The system is also scalable, meaning it can be expanded by adding more sensors or nodes. This allows the system to be adapted for larger mining operations.

Overall, the performance analysis indicates that the proposed system meets the requirements of a reliable and efficient safety monitoring solution.

4.6 Discussion

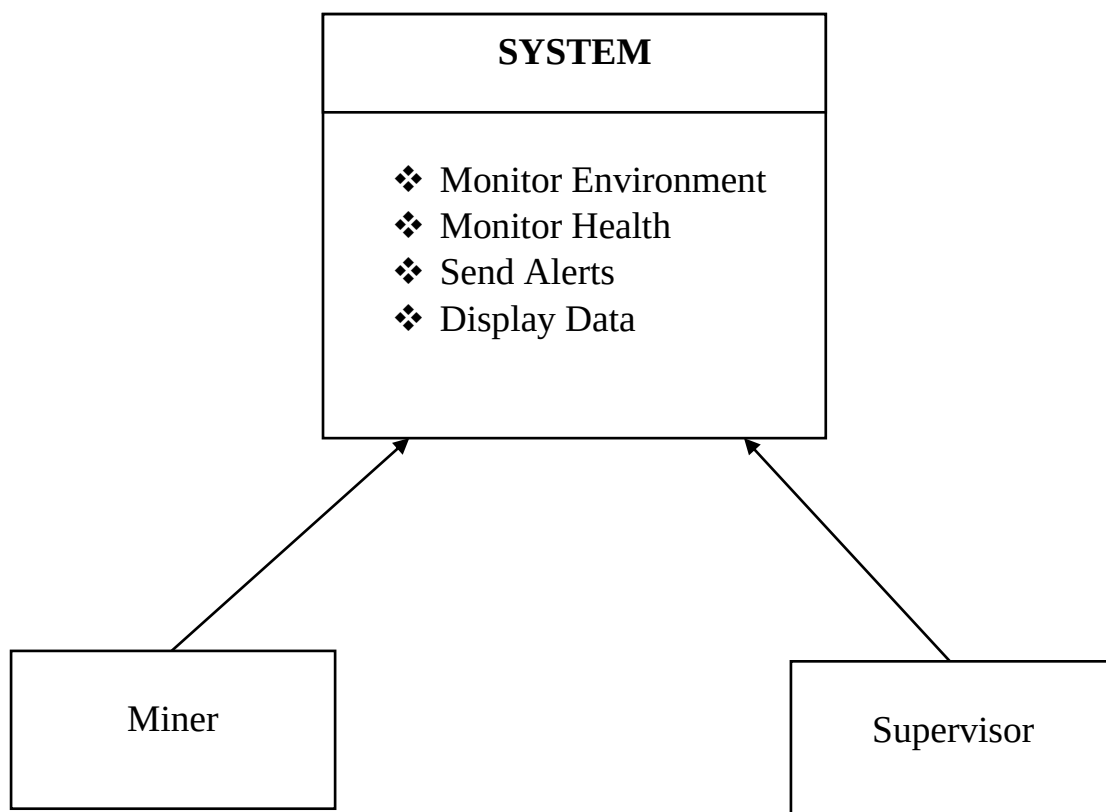


Fig 4.6 The discussed system mechanism

The integration of multiple technologies, including IoT, wireless communication, and computer vision, provides a comprehensive solution for monitoring and control.

The hardware prototype demonstrates the feasibility of deploying the system in real-world conditions. The wearable design ensures ease of use for miners, while the compact structure makes it practical for everyday use.

The results obtained from the implementation of the proposed system clearly indicate its effectiveness in improving safety in coal mining environments.

The dashboard provides a centralized platform for monitoring all aspects of the system, including environmental conditions, worker status, and alerts. This enables supervisors to make quick and informed decisions, thereby reducing the risk of accidents.

The inclusion of computer vision for attendance monitoring adds an innovative feature to the system. It not only improves safety but also enhances management efficiency.

However, there are certain areas where improvements can be made. The accuracy of sensors can be further enhanced, and additional features such as predictive analytics can be implemented to anticipate potential hazards.

Despite these limitations, the proposed system provides a reliable, efficient, and cost-effective solution for coal mine safety. It demonstrates the potential of IoT-based technologies in transforming traditional safety systems into smart and intelligent solutions.

The proposed system provides an effective solution for monitoring worker health and environmental conditions in real time. It enables early detection of

hazardous situations and helps in reducing the risk of accidents, thereby improving overall safety in coal mining environments.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The developed Smart Safety System for Coal Mines demonstrates an effective approach for improving worker safety and monitoring in underground mining environments. Mining industries often face critical challenges due to hazardous gases, high temperatures, poor visibility, and limited communication facilities. To address these issues, the proposed system combines IoT technology, wireless communication, and real-time monitoring into a single integrated platform capable of supporting safer mining operations.

The implemented system continuously monitors important environmental and worker health parameters using multiple sensors. The gas sensor helps identify the presence of harmful gases inside the mining area, while the DHT22 sensor measures surrounding temperature conditions. In addition, the MAX30102 sensor is used to monitor the heart rate of workers, allowing the system to observe their physical condition during operation. These parameters are processed using the ESP32 controller and transmitted through LoRa communication, which provides reliable long-range data transfer suitable for underground conditions where conventional wireless communication becomes unreliable.

The receiver section, based on the ESP8266 module, acts as an internet gateway by forwarding the received information to a web-based dashboard through Wi-Fi connectivity. The dashboard enables supervisors to observe real-time data remotely and provides a centralized monitoring platform for safety management. The integration of graphical monitoring features improves visualization and helps in quick decision-making during emergency situations.

The project also introduces additional safety-oriented concepts such as BLE-based safety equipment verification and computer vision-based attendance monitoring. These features improve worker management and encourage adherence to safety protocols. Although the computer vision module was not fully integrated into the final implementation, the proposed design demonstrates the possibility of expanding the system into a more intelligent and automated monitoring platform in the future.

The experimental results indicate that the proposed system is capable of stable sensor monitoring, reliable wireless communication, and real-time data transmission. Overall, the project highlights the importance of combining sensing technologies, wireless communication, cloud connectivity, and intelligent monitoring techniques to build modern industrial safety systems. The proposed solution represents a practical step toward the development of safer, smarter, and more efficient mining operations.

5.2 Future Scope

The proposed system can be further enhanced by integrating advanced technologies to improve its performance, reliability, and intelligence. One possible improvement is the implementation of artificial intelligence and machine learning algorithms for predictive safety analysis. By analyzing sensor data continuously, the system can predict hazardous conditions before accidents occur and provide early warnings to workers and supervisors.

The computer vision module can also be expanded in future work by integrating real-time PPE detection using deep learning models such as YOLO or TensorFlow-based object detection systems. This would enable automatic verification of helmets, safety vests, and masks without manual supervision.

Another important enhancement is the development of a multi-node monitoring architecture capable of supporting large-scale mining environments. Multiple sensor nodes and camera units can be interconnected through LoRa networks to provide complete underground coverage and centralized monitoring.

The dashboard can be further improved by adding cloud storage, mobile application support, emergency alert notifications, and advanced graphical analysis tools. This would allow supervisors to access real-time data and historical records from any location.

Future versions of the system can also focus on improving power efficiency by using low-power hardware components and optimized communication techniques. This would increase battery life and make the wearable safety unit more practical for long-duration mining operations.

In addition, integrating edge AI devices and automated emergency response mechanisms can significantly improve system response time and reliability. The overall system has strong potential for real-world deployment and can be adapted for use in other hazardous industrial environments such as chemical plants, construction sites, and oil refineries

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